

Notes: Petersen and Rajan (JF, 2002)

Does Distance Still Matter? The Information Revolution in Small Business Lending

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Main datasets and sample construction	4
2.2	Unit of observation and market objects	4
2.3	Distance and communication measures	5
2.4	Key descriptive facts	5
2.5	Unobservables, timing, and synthetic-panel issues	6
3	Models and empirical specifications (all equations + interpretation)	6
3.1	Baseline distance regression	6
3.2	Communication specification	7
3.3	Loan-type heterogeneity	7
3.4	Within-firm and transparency specifications	7
3.5	Alternative explanations: consolidation, deposits, and credit quality	8
3.6	Technology and productivity mechanism	9
3.7	Predicted distance and predicted communication as transparency measures	9
3.8	Consequences for loan pricing and approval	10
4	Identification and instruments (what makes the estimates credible)	10
4.1	What fails in naive interpretations	10
4.2	What identifies the central result	11
4.3	Why the alternative explanations fail	11
4.4	Why the information-technology story is credible	11
4.5	Relation to the literature	12

5	Tables and figures	12
5.1	Table 1: raw changes in distance and communication	12
5.2	Table 2: baseline distance regressions	12
5.3	Table 3 and Table 4: loan-type robustness, fixed effects, and transparency	13
5.4	Figure 1 and Figure 2: sample-comparison check and fitted distance trend	13
5.5	Table 5 and Figure 3: communication becomes impersonal	15
5.6	Table 6, Table 7, Figure 4, and Figure 5: why consolidation is not the answer, and productivity is	15
5.7	Table 8 and Table 9: credit-quality consequences	17
6	Short summary	17
7	Conclusion	18
7.1	Core conclusion (what the paper establishes)	18
7.2	Economic intuition	18
7.3	Takeaway	18

1 Big picture

- **Question.** Does geographic distance still matter in small business lending, or has the information revolution made it possible for lenders to lend effectively at a distance?
- **Setting.** The paper studies U.S. small business lending relationships formed between **1973 and 1993**, using the **1993 National Survey of Small Business Finance (NSSBF)** and, where possible, the **1988 NSSBF**. The focus is on the physical distance between a firm and its lender and on whether they communicate **in person**, **by phone**, or **by mail**.
- **Core challenge.** Classic relationship-lending theories stress that small business lending depends on **soft information**, which is hard to transmit and therefore should make lending local, personal, and branch-based. If the observed rise in lending distance is real, the paper must explain why it happened and why it did not generate a collapse in loan quality.
- **Main contribution.** The paper documents two new secular facts:
 1. the physical distance between small firms and their lenders rose sharply over time;
 2. borrower–lender communication became much less personal over time.

It then shows that these patterns are **not** mainly explained by firm composition, bank consolidation, urbanization, deposit-market deregulation, or relaxed credit standards. Instead, the evidence points to **improvements in lender productivity and information technology**.

- **Why it matters.** If small business lending is becoming less local, then the relevant geographic market for credit is expanding. That matters for **bank-merger policy**, **antitrust market definition**, **local financial stability**, and the extent to which **small firms are financially constrained**.
- **Main empirical question.** Are small firms actually borrowing from more distant lenders and communicating in less personal ways, and if so, is this because the lending technology itself changed?
- **Main empirical results.**
 - The average distance to a lender rises from about **51 miles** for relationships begun in the 1970s to about **161 miles** for those begun in the 1990s.
 - The median distance rises from **3 miles** to **15 miles**.
 - The fraction of relationships conducted predominantly **in person** falls from about **68%** in the 1970s to about **34%** in the early 1990s.
 - In baseline regressions, the log distance to the lender grows by about **3.4% per year**, even after controlling for firm, loan, and lender characteristics.
 - The probability of communicating in person falls from about **59%** for a relationship started in 1973 to about **36%** for one started in 1993 after controls.
 - Bank consolidation does **not** explain the pattern: the number of banks falls only from the mid-1980s onward, while branch counts do not decline in the same way.
 - Measures of **bank labor productivity** explain a large share of the distance and communication trends; once these are included, the pure time trend becomes small or insignificant.

- Firms that are predicted to be able to borrow at a distance because they are more informationally transparent get **lower interest rates** and **higher approval probabilities**; however, the approval advantage of such transparency weakens over time, suggesting that distant credit becomes available to a broader set of firms.
- **Economic takeaway.** The paper argues that information technology has weakened the “tyranny of distance” in small business lending. Hard information, centralized databases, and automated credit evaluation have partially substituted for soft-information-intensive local lending relationships.

2 Data and measurement (key objects)

2.1 Main datasets and sample construction

- **Primary source.** The core data come from the **1993 National Survey of Small Business Finance**. This is a stratified random sample collected by the Federal Reserve and the Small Business Administration.
- **Supplementary source.** The paper supplements the 1993 survey with the **1988 NSSBF** whenever the relevant variables are available. This helps address sample-selection concerns from the synthetic-panel design.
- **Firm definition.** To enter the survey, a firm must be **for-profit** and have fewer than **500 employees**. So the entire analysis is about genuinely small firms.
- **Firm size in the sample.**
 - Mean 1992 sales: about **\$3.6 million**; median: **\$400,000**.
 - Mean book assets: about **\$1.7 million**; median: **\$153,000**.
- **Relationship history.** Firms report an exhaustive list of financial institutions with which they have a relationship and how long they have been dealing with each institution. That allows the authors to recover the **calendar year in which a lending relationship began**.

2.2 Unit of observation and market objects

- **Unit of observation.** The basic unit is a **firm–lender–loan-type pair**. If a firm borrows from multiple lenders, or from the same lender through multiple loan types (e.g., line of credit and mortgage), it generates multiple observations.
- **Baseline sample size.** The main distance sample contains about **5,981** firm–lender pairs formed between **1973 and 1993**. The expanded 1988+1993 sample has about **9,385** observations.
- **Types of financial relationships.** The survey records credit, service, and deposit relationships. The paper focuses on **credit relationships** but also uses information on transaction relationships such as checking accounts.
- **Lender categories.**
 - **Banks:** commercial banks, savings banks, and credit unions.

- **Nonbank financial lenders:** finance companies, insurance companies, brokerage firms, leasing companies, mortgage banks, venture capitalists.
- **Nonfinancial lenders:** mostly suppliers and other trade creditors.
- **Debt shares in the sample.** Of external debt, roughly **69%** is from banks, **25%** from nonbank financial lenders, and about **5%** from nonfinancial lenders.

2.3 Distance and communication measures

- **Distance.** Distance is measured in miles from the firm's main office to the office or branch of the lender that the firm uses most frequently.
- **Why log distance.** The distribution of raw distance is highly skewed, so the preferred measure is

$$\log(1 + \text{Distance}_{ilr}).$$

This reduces skewness and fits the idea that the economic difference between 5 and 100 miles is much larger than the difference between 1,005 and 1,100 miles.

- **Communication.** Firms report the predominant method of communication with each lender: **in person**, **by phone**, or **by mail**.

2.4 Key descriptive facts

- **Overall distance levels.**
 - Mean distance overall: about **115 miles**.
 - Median distance overall: about **9 miles**.
- **Distance by decade of relationship start.**
 - 1973–1979: mean **51.2** miles, median **3.0** miles.
 - 1980–1989: mean **92.6** miles, median **7.0** miles.
 - 1990–1993: mean **161.3** miles, median **15.0** miles.
- **Banks versus nonbanks.**
 - Bank relationships remain much closer than nonbank relationships, but even banks move out sharply over time: mean bank distance rises from **15.8** to **67.8** miles.
 - Nonbanks are much farther away throughout, with mean distances above **200 miles**.
- **Communication by decade.**
 - For all lenders combined, the share of relationships conducted predominantly in person falls from **0.68** in 1973–1979 to **0.34** in 1990–1993.
 - For banks, the in-person share falls from **0.77** to **0.54**.
 - Nonbanks are much more impersonal even in the 1970s.

2.5 Unobservables, timing, and synthetic-panel issues

- **Synthetic panel.** The paper reconstructs relationship cohorts from cross-sectional surveys by using the reported year a relationship began.
- **Selection issue.** The 1993 survey only includes firms and relationships that survive until 1993. This creates a possible survival bias that the authors explicitly test using the 1988 survey.
- **Unobserved firm heterogeneity.** The paper addresses persistent unobserved firm traits using **firm fixed effects**, between-firm versus within-firm decompositions, and direct controls for transparency, ownership, and sales scope.
- **Credit quality proxies.** The paper does not observe historical credit quality at relationship start. Instead, it later uses the **interest rate on the most recent loan** and whether the firm's **most recent loan application was approved** as observable proxies for current credit quality and financing access.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline distance regression

The main reduced-form specification is

$$\log(1 + \text{Distance}_{ilr}) = \alpha + \beta \text{YearStart}_{ilr} + X'_{ilr}\gamma + \varepsilon_{ilr},$$

where:

- i indexes the firm,
- l indexes the lender,
- r indexes the loan relationship or loan type,
- YearStart_{ilr} is the calendar year in which the relationship began,
- X_{ilr} includes borrower, lender, and loan controls.

Interpretation. The coefficient β is the central summary of the paper. A positive β means that newly formed relationships are systematically more distant than older cohorts of relationships, after controlling for observable composition.

- In the baseline 1993 sample, $\hat{\beta} \approx 0.034$, meaning distance grows about **3.4% per year**.
- In the 1988+1993 expanded sample, $\hat{\beta} \approx 0.038$.
- In weighted estimates, $\hat{\beta} \approx 0.033$.
- In the subsample where the first lender appears within five years of firm birth, $\hat{\beta} \approx 0.045$.

3.2 Communication specification

The paper estimates ordered-logit and binary-logit models for the method of communication. In the binary version,

$$\Pr(\text{Impersonal}_{ilr} = 1) = \Lambda(\alpha + \beta \text{YearStart}_{ilr} + X'_{ilr}\gamma),$$

where $\text{Impersonal} = 1$ if communication is by phone or by mail, and 0 if it is in person.

The paper also augments this with actual distance:

$$\Pr(\text{Impersonal}_{ilr} = 1) = \Lambda(\alpha + \beta \text{YearStart}_{ilr} + \delta \log(1 + \text{Distance}_{ilr}) + X'_{ilr}\gamma).$$

Interpretation.

- A positive year coefficient means newer relationships are less personal.
- Including actual distance asks whether communication became less personal **over and above** the fact that lenders became farther away.

Empirically:

- Without controlling for actual distance, the year coefficient is about **0.046**.
- After adding actual distance, it falls to about **0.031** but remains strongly significant.

So the shift toward phone and mail communication is **not just** a mechanical consequence of longer geographic distance.

3.3 Loan-type heterogeneity

To see whether the distance trend is just a change in the mix of loans, the paper runs separate regressions by loan type:

- lines of credit,
- leases,
- mortgages,
- motor vehicle loans,
- equipment loans,
- other loans.

Interpretation. If the year coefficient is positive across loan categories, then the distance trend is not simply due to a shift from relationship-heavy loans to standardized loans.

Empirically, the year coefficient is positive in every category, ranging roughly from **0.028** to **0.055**. This is strong evidence against a pure loan-composition story.

3.4 Within-firm and transparency specifications

The paper next estimates versions of the distance regression with:

1. **industry controls**,
2. **firm fixed effects** (within-firm variation),
3. **between-firm means**,
4. direct controls for **information transparency**.

A conceptual version is

$$\log(1 + \text{Distance}_{ilr}) = \alpha_i + \beta \text{YearStart}_{ilr} + X'_{ilr}\gamma + Z'_i\delta + \varepsilon_{ilr},$$

where α_i is a firm fixed effect and Z_i captures firm transparency.

The transparency variables include:

- whether the firm had financial records available when answering the survey,
- whether it has a business credit card,
- ownership concentration,
- corporate form,
- franchise status,
- owner management,
- owner age and experience,
- regional or national sales area.

Interpretation. More transparent firms should be able to borrow farther away because hard information about them travels better.

Key results:

- The year coefficient remains positive in the within-firm specification (about **0.040**).
- Firms with records are about **9%** farther from lenders.
- Franchises are about **29%** farther.
- Going from fully diffuse to fully concentrated ownership reduces distance by about **26%**.
- Firms with national sales areas borrow substantially farther away.

3.5 Alternative explanations: consolidation, deposits, and credit quality

The paper tests several alternative explanations by augmenting the basic regressions with variables that change over time and region.

A generic specification is

$$\log(1 + \text{Distance}_{ilr}) = \alpha + \beta \text{YearStart}_{ilr} + X'_{ilr}\gamma + M'_{ct}\theta + \varepsilon_{ilr},$$

where M_{ct} is a vector of lending-environment variables for census region c and time t .

The paper considers:

- bank density,
- branch density,
- loan losses/total loans,
- bank employees per unit of real lending,
- bank employees per unit of regional real GDP.

Interpretation.

- If consolidation is the main story, bank density or branch density should explain away the time trend.
- If lax credit standards are the story, loan losses should trend up and explain the change.
- If information technology is the story, measures of lender productivity should explain a large part of the trend.

The results strongly favor the last interpretation.

3.6 Technology and productivity mechanism

The authors interpret the declining ratio of bank employees to loan volume or regional output as a proxy for greater use of information technology and automated screening/monitoring.

In the distance regression, the key specification is conceptually:

$$\log(1 + \text{Distance}_{ilr}) = \alpha + \beta \text{YearStart}_{ilr} + X'_{ilr}\gamma + \theta \log\left(\frac{\text{Bank Employees}_{ct}}{\text{Real GDP}_{ct}}\right) + \varepsilon_{ilr}.$$

Interpretation. If fewer employees are needed per dollar of regional activity, lending technology is more productive. That should allow more distant lending.

Empirically:

- A **10% decrease** in bank employees relative to real regional GDP increases lender distance by about **4.7%**.
- This explains roughly **58%** of the predicted increase in distance over the period.
- Once this productivity measure is included, the pure year trend becomes small and statistically weak.

The same logic holds for communication: once bank productivity measures are included, the year effect on impersonal communication becomes insignificant.

3.7 Predicted distance and predicted communication as transparency measures

Actual distance is a noisy measure of informational transparency because a firm may be far from its lender simply because there are no nearby lenders. To isolate the transparency component, the paper constructs:

$$\hat{D}_i = Z'_i \hat{\delta}, \quad \hat{C}_i = Z'_i \hat{\eta},$$

where Z_i includes only firm characteristics that proxy for how observable or standardized the firm is.

Interpretation. These fitted values measure the firm's **ability to borrow at a distance or communicate impersonally because of public/transmissible information**, not because of lender scarcity.

3.8 Consequences for loan pricing and approval

The paper studies whether informationally transparent firms get better credit terms and whether this changes over time.

For rates:

$$\text{Rate}_i = \alpha + \beta \hat{D}_i + \gamma(\hat{D}_i \times \text{YearStart}_i) + X_i' \theta + u_i.$$

For approval:

$$\Pr(\text{Approved}_i = 1) = \Lambda\left(\alpha + \beta \hat{D}_i + \gamma(\hat{D}_i \times \text{YearStart}_i) + X_i' \theta\right).$$

The same exercise is repeated using predicted method of communication instead of predicted distance.

Interpretation.

- If transparency is valuable, it should reduce loan rates and improve approval odds.
- If improved monitoring technology made distant lending feasible for a wider group of firms, the **credit-quality signal** embedded in distance/transparency should weaken over time.

Results:

- Predicted distance significantly lowers interest rates: a **50% increase** in predicted distance lowers borrowing cost by about **25 basis points**.
- But the interaction of predicted distance with year is **not significant** in the rate regression.
- Predicted distance strongly raises the probability of approval; moving from the 25th to the 75th percentile raises approval probability by about **21 percentage points**.
- The interaction between predicted distance and year is **negative and significant** in the approval regression, implying that the advantage of transparency weakens over time.
- The implied credit-quality signal of distance is about **25% weaker in 1993 than in 1973**.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in naive interpretations

- **Raw averages are not enough.** The rise in mean distance could come from changes in lender type, loan type, urbanization, or firm composition.
- **Synthetic-panel bias is a concern.** Because the paper reconstructs relationship histories from cross-sectional surveys, one might worry that only long-lived firms or long-lived firm-lender pairs remain in the sample.

- **Bank consolidation is an obvious suspect.** Fewer banks might mechanically force firms to go farther.
- **Lax underwriting is another suspect.** If lenders simply became careless, more distant lending would not indicate true technological change.

4.2 What identifies the central result

- **Rich observable controls.** The baseline regressions control for firm age, lender type, checking-account relationships, whether the lender is the first lender, collateralization, and later many firm-specific characteristics.
- **Expanded 1988+1993 sample.** The 1988 survey provides a direct check on whether survival bias is driving the result. The distance trend remains and even strengthens slightly.
- **Within-firm variation.** The firm fixed-effects specification shows that the trend does not come only from changes in the cross-section of firms.
- **Loan-type robustness.** Positive year coefficients across all major loan categories reject the idea that the trend is just a change in loan mix.

4.3 Why the alternative explanations fail

- **Changing lender mix explains only part of the raw increase.** Yes, relationships move somewhat from close bank/checking-account lenders toward more distant lenders, but even conditional on lender type, distance rises sharply.
- **Deposit-market deregulation is too small a channel.** Distance to checking-account providers rises at only about **1.4% per year**, compared with about **3.4% per year** for lending distance.
- **Bank consolidation timing does not fit.** The sharp fall in the number of banks begins in the mid-1980s, but the distance trend begins much earlier.
- **Branch counts do not collapse.** Physical branches rise for much of the period, so firms are not simply being left without local offices.
- **Loan losses do not support a weakening-standards story.** Loan-loss variables are statistically unimportant, and there is no secular increase in loan losses that mirrors the distance trend.

4.4 Why the information-technology story is credible

- **Hard information becomes more available.** The paper points to Dun & Bradstreet and other infomediaries. D&B coverage of firms grows about **6.3% per year**, much faster than real economic growth.
- **Automated appraisal fits the institutional details.** Credit scoring, computerized reports, and analytic rules lower the labor intensity of origination and monitoring.

- **Regional productivity measures line up with the timing and geography of the trend.** Bank employees relative to regional output, not simply national trends, explain the local evolution of distance and communication.
- **The final quality test is consistent with the mechanism.** Distant borrowing becomes a weaker indicator of top borrower quality over time, which is exactly what one would expect if technology allowed lenders to lend farther out in the quality distribution.

4.5 Relation to the literature

- The paper extends the relationship-lending literature of **Sharpe (1990)**, **Rajan (1992)**, and **Petersen and Rajan (1994, 1995)** by showing how the geography of small business lending itself changes.
- It connects to work on soft versus hard information, later formalized by **Stein (2002)** and summarized by **Liberti and Petersen (2019)**.
- It also contributes to debates on bank consolidation, local market power, and the real effects of financial technology.

5 Tables and figures

5.1 Table 1: raw changes in distance and communication

- **Read:** This is the paper’s basic descriptive fact table.
- **Distance.** Mean distance rises from **51.2 miles** in relationships begun in the 1970s to **161.3 miles** in those begun in the 1990s; the median rises from **3** to **15** miles.
- **By lender type.** Banks remain much closer than nonbanks, but even bank distance rises strongly over time.
- **Communication.** The share of in-person communication falls from **0.68** to **0.34** overall.
- **Punchline.** The raw trends are large, economically meaningful, and visible before any regression analysis.

5.2 Table 2: baseline distance regressions

- **Read:** This table shows that the time trend in distance survives controls.
- **Key coefficient.** The baseline coefficient on year relation started is about **0.034**; in the expanded sample it is about **0.038**; in weighted regressions it is about **0.033**.
- **Lender type.** Banks are much closer than nonbanks, and checking-account banks are especially close.
- **Sample-selection test.** The expanded 1988+1993 sample does not make the trend disappear.
- **Punchline.** The rise in lending distance is not an artifact of obvious composition effects.

Table I
Firm's Lending Relationship

The table is based on data from the 1993 National Survey of Small Business Finance. Each observation represents a firm/lender/loan type pair. Firms that borrow from multiple lenders or that borrow from a given lender but through multiple loan types (e.g., a line of credit and a mortgage) will generate multiple observations. The sample contains 5,981 pairs, which were begun between 1973 and 1993. Panel A contains the distance between a firm and its lender classified by lender type and the year the relationship began. The first entry in each cell is the mean distance, the second entry is the median distance, and the third entry is the 75th percentile. The far right column contains the sample average by lender types, and the bottom row contains the sample average by decade. Distance is measured from the main office of the firm to the office or branch of the lender which the firm uses most frequently and is reported in miles. Banks include commercial banks, savings banks, and credit unions. Nonbanks include finance companies, insurance companies, brokerage firms, leasing companies, mortgage banks, and venture capitalists. Panel B contains data on the predominant method of communication between the firm and its lender. The communication can be in person (1), by phone (2) or by mail (3). The first number is the fraction of firm-lender pairs that communicate predominantly in person. The second number is the average value of the method of communication variable. Higher values are associated with less personal communication.

Lender type	Year Lending Relationship Began			
	1973-1979	1980-1989	1990-1993	Total
Panel A: Distance to a Firm's Lender by Lender Type and Time				
Banks	15.8 2.0 6.0	34.0 4.0 12.0	67.8 5.0 20.0	42.5 4.0 14.0
Nonbanks	235.9 15.5 71.0	222.1 42.0 215.5	280.5 54.0 332.0	251.6 45.0 255.0
Nonfinancial firms	117.3 17.5 60.0	165.9 29.0 141.5	209.2 32.0 235.0	182.5 30.5 164.0
Total	51.2 3.0 10.0	92.6 7.0 33.0	161.3 15.0 91.5	114.7 9.0 42.0
Panel B: Method of Communication to a Firm's Lender by Lender Type and Time				
Banks	0.77 1.3	0.67 1.5	0.54 1.7	0.64 1.5
Nonbanks	0.27 2.2	0.12 2.5	0.09 2.6	0.11 2.6
Nonfinancial firms	0.35 2.2	0.20 2.4	0.18 2.4	0.20 2.4
Total	0.68 1.5	0.49 1.8	0.34 2.1	0.46 1.9

Table II
Determinants of Distance to the Firm's Lenders

The table contains regressions where the dependent variable is the log of one plus the distance to the firm's lender. The sample includes only relationships beginning since 1973. Dummy variables are included for whether the lender is a bank or a nonfinancial firm. The missing category is nonbank financial lenders. Each observation represents a firm-lender pair. Firms that borrow from multiple lenders or that borrow from a given lender but through multiple loan types (e.g., a line of credit and a mortgage) will generate multiple observations. The data are taken from the 1993 National Survey of Small Business Finance only in column I. We include data from the 1988 National Survey of Small Business Finance in the remaining columns. The models are described as follows: I. Basic regression model. II. Expanded sample. Sample includes observations from the 1993 and the 1988 National Survey of Small Business Finance. III. Expanded sample. These estimates are based on a weighted least squares, where the population weights from the surveys are used. IV. Expanded sample. Sample includes only observations for firms where the first reported lending relationship is within five years of the firm's birth. V. Expanded sample. Lines of credit: The regression includes both a dummy variable for whether the loan is a line of credit and an interaction between the line of credit dummy and the year the relationship started. This allows the slope on year to vary by whether the loan is a line of credit or not.

Independent Variables	Models				
	I	II	III	IV	V
Firm's age	0.001 (0.002)	0.003 ¹⁰ (0.001)	-0.001 (0.002)	-0.005 (0.004)	0.002 ¹⁰ (0.001)
Year relation started	0.034 ¹ (0.005)	0.038 ¹ (0.004)	0.033 ¹ (0.004)	0.045 ¹ (0.005)	0.041 ¹ (0.005)
Lender is bank	-1.173 ¹ (0.062)	-1.176 ¹ (0.050)	-1.310 ¹ (0.047)	-1.190 ¹ (0.061)	-1.192 ¹ (0.050)
Lender is a nonfinancial firm	-0.369 ¹ (0.097)	-0.357 ¹ (0.077)	-0.493 ¹ (0.073)	-0.421 ¹ (0.090)	-0.347 ¹ (0.077)
Lender provides a checking account	-1.062 ¹ (0.060)	-0.990 ¹ (0.047)	-1.057 ¹ (0.043)	-1.031 ¹ (0.057)	-1.043 ¹ (0.048)
Lender is firm's first lender	-0.163 ¹ (0.049)	-0.203 ¹ (0.039)	-0.198 ¹ (0.038)	-0.190 ¹ (0.054)	-0.200 ¹ (0.039)
Loan is collateralized	0.081 (0.062)	0.132 ¹ (0.049)	0.041 (0.045)	0.081 (0.058)	0.256 ¹ (0.054)
Observation from 1993 sample		-0.097 ⁵ (0.039)	0.018 (0.037)	-0.044 (0.046)	-0.123 ¹ (0.040)
LOC					0.369 ¹ (0.098)
LOC * year relation started					-0.011 (0.007)
R ²	0.308	0.296	0.335	0.307	0.298
Number of observations	5,981	9,385	9,385	6,681	9,385

¹⁰, ⁵, ¹ Significance at the 10%, 5%, and 1% levels, respectively.

5.3 Table 3 and Table 4: loan-type robustness, fixed effects, and transparency

- **Read:** Table 3 shows the time trend exists across all major loan categories. Table 4 shows it also survives industry controls, firm fixed effects, and direct measures of informational transparency.
- **Important magnitudes.**
 - Loan-type year coefficients are positive across the board.
 - Within-firm year coefficient is about **0.040**.
 - Firms with records are about **9%** farther from lenders.
 - Franchises are about **29%** farther.
 - National-sales firms are substantially farther from lenders.
- **Punchline.** More transparent firms can borrow from farther away, which supports the hard-information mechanism.

5.4 Figure 1 and Figure 2: sample-comparison check and fitted distance trend

- **Figure 1.** The 1988 and 1993 surveys both show an upward relation between distance and relationship-start year. This visually weakens the claim that sample selection is generating the result.
- **Figure 2.** The fitted log distance rises steadily across relationship cohorts even after rich controls.
- **Punchline.** The trend is smooth, long-run, and not confined to a late-period break.

Table III

Determinants of Distance to the Firm's Lenders by Loan Type

The table contains regressions where the dependent variable is the log of one plus the distance to the firm's lender. A separate regression was run for each of the six loan types in the sample: lines of credit (column I), leases (column II), mortgages (column III), motor vehicle loans (column IV), equipment loans (column V), and other loans (column VI). The sample includes only relationships beginning since 1973. Dummy variables are included for whether the lender is a bank or a nonfinancial firm. The missing category is nonbank financial lenders. Each observation represents a firm-lender pair. Firms that borrow from multiple lenders or that borrow from a given lender but through multiple loan types (e.g., a line of credit and a mortgage) will generate multiple observations. The data are taken from the 1988 and 1993 National Survey of Small Business Finance.

Independent Variables	I	II	III	IV	V	VI
Firm's age	0.001 (0.002)	0.005 (0.004)	0.011 ⁵ (0.005)	-0.005 (0.003)	0.002 (0.004)	0.007 (0.005)
Year relation started	0.028 ¹ (0.009)	0.055 ¹ (0.015)	0.042 ¹ (0.011)	0.035 ¹ (0.008)	0.035 ¹ (0.011)	0.037 ¹⁰ (0.019)
Lender is bank	-1.410 ¹ (0.132)	-0.870 ¹ (0.189)	-1.472 ¹ (0.182)	-0.954 ¹ (0.079)	-1.083 ¹ (0.138)	-1.210 ¹ (0.202)
Lender is a nonfinancial firm	-0.250 (0.242)	-0.491 ¹ (0.172)	-0.548 (0.369)	-0.055 (0.198)	-0.763 ¹ (0.149)	-0.661 ⁵ (0.261)
Lender provides a checking account	-1.019 ¹ (0.088)	-1.238 ¹ (0.245)	-0.662 ¹ (0.115)	-0.910 ¹ (0.088)	-1.538 ¹ (0.133)	-1.371 ¹ (0.147)
Lender is firm's first lender	-0.175 ⁵ (0.070)	-0.240 ⁵ (0.117)	-0.385 ¹ (0.125)	-0.080 (0.071)	-0.213 ⁵ (0.101)	-0.182 (0.145)
Loan is collateralized	0.248 (0.139)					-0.015 (0.300)
Year relation started if loan is collateralized	0.005 (0.011)					0.018 (0.024)
Observation from 1993 sample	-0.154 ⁵ (0.068)	-0.210 (0.138)	-0.257 ⁵ (0.113)	-0.187 ¹ (0.073)	-0.035 (0.105)	0.940 (0.144)
R ²	0.233	0.138	0.211	0.246	0.349	0.311
Number of observations	2,559	1,329	976	2,309	1,481	731
Median distance	5.0	41.0	6.0	11.0	11.0	6.0

¹⁰, ⁵, ¹ Significance at the 10%, 5%, and 1% levels, respectively.

Table IV

Determinants of Distance to the Firm's Lenders: Measuring the Informational Transparency of Firms

The table contains regressions where the dependent variable is the log of one plus the distance to the firm's lender. The sample includes only relationships beginning since 1973. Each observation represents a firm-lender pair. Firms that borrow from multiple lenders or that borrow from a given lender but through multiple loan types (e.g., a line of credit and a mortgage) will generate multiple observations. The data are taken from the 1988 National Survey of Small Business Finance, except for column V, which also includes data from the 1993 National Survey of Small Business Finance. Models are described as follows: I. Industry controls. The regression includes dummy variables to control for differences across the 50 two-digit industries in the sample. II. Within estimates. A dummy variable is included for every firm. Thus, the coefficients are estimates based on variation from firm specific means. The R² includes the explanatory power of the firm effects. Without including the explanatory power of the firm effects, the R² would be 0.307. III. Between estimates. The coefficients are estimated based on variation between firm specific means. Each observation represents the mean value for a given firm. Thus, the sample size is reduced to the number of unique firms in our sample. IV. Firm-specific variables. This model includes additional controls for firm-specific characteristics. V. Expanded sample. This model includes data from both the 1988 and 1993 National Survey of Small Business Finance. Only variables that are available in both surveys are included in the regression. VI. Firm-specific variables. This model also includes the log of the firm's total assets.

Independent Variables	Models					
	I	II	III	IV	V	VI
Firm's age	0.001 (0.002)		0.000 (0.002)	0.000 (0.002)	0.002 (0.001)	-0.000 (0.002)
Year relation started	0.033 ¹ (0.005)	0.040 ¹ (0.011)	0.035 ¹ (0.006)	0.037 ¹ (0.005)	0.040 ¹ (0.004)	0.042 ¹ (0.005)
Lender is bank	-1.164 ¹ (0.062)	-1.121 ¹ (0.085)	-1.102 ¹ (0.089)	-1.152 ¹ (0.082)	-1.163 ¹ (0.080)	-1.140 ¹ (0.062)
Lender is a nonfinancial firm	-0.329 ¹ (0.096)	-0.124 (0.132)	-0.472 ¹ (0.139)	-0.373 ¹ (0.096)	-0.354 ¹ (0.077)	-0.356 ¹ (0.096)
Lender provides a checking account	-1.079 ¹ (0.059)	-1.063 ¹ (0.083)	-1.116 ¹ (0.081)	-1.106 ¹ (0.080)	-1.071 ¹ (0.047)	-1.137 ¹ (0.060)
Lender is firm's first lender	-0.143 ¹ (0.048)	-0.026 (0.077)	-0.258 ¹ (0.085)	-0.133 ¹ (0.049)	-0.166 ¹ (0.039)	-0.095 ¹⁰ (0.062)
Loan is collateralized	0.028 (0.062)	0.011 (0.088)	0.022 (0.061)	0.059 (0.062)	0.118 ⁵ (0.049)	0.037 (0.062)
Firm had records for filling out survey				0.091 ¹ (0.042)	0.131 ¹ (0.034)	0.054 ¹ (0.043)

Firm has credit card	0.003 (0.043)					-0.000 (0.042)
Ownership share of largest owner (%)	-0.255 ¹ (0.076)					-0.151 ¹⁰ (0.076)
Corporation (1 = yes)	0.066 (0.052)	0.195 ¹ (0.038)				-0.006 (0.054)
Franchise (1 = yes)	0.250 ¹ (0.091)					0.242 ¹ (0.091)
Owner managed	-0.044 (0.050)	-0.076 ¹⁰ (0.044)				-0.008 (0.051)
Owner's age when relation began	0.000 (0.003)					0.000 (0.003)
Owner's experience when relation began	-0.003 (0.003)					-0.006 (0.003)
Sales area regional (1 = yes)	0.073 (0.047)					0.040 (0.047)
Sales area national (1 = yes)	0.359 ¹ (0.060)					0.292 ¹ (0.061)
Firm in MSA (1 = yes)	-0.127 ¹ (0.056)	-0.126 ¹ (0.042)				-0.137 ¹ (0.055)
Herfindahl Index > 1,500	0.046 (0.045)	0.036 (0.037)				0.045 (0.045)
Observation from 1993 sample		-0.080 ¹⁰ (0.041)				
Log(firm assets)						0.069 ¹ (0.013)
Industry controls F-statistics (p-value)	2.993 (0.000)					
Firm controls F-statistics (p-value)		1.502 (0.000)				
R ²	0.327	0.397	0.318	0.301	0.319	0.319
Number of observations	5,981	5,981	2,878	5,974	9,378	5,974

¹⁰, ⁵, ¹ Significance at the 10%, 5%, and 1% levels, respectively.

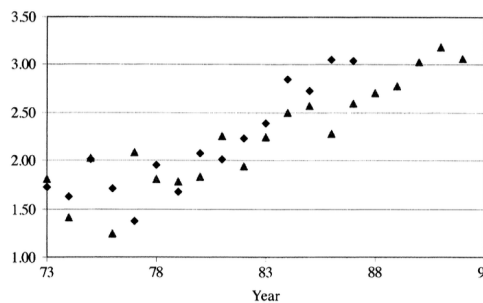


Figure 1. Changes in the average distance to lender over time by sample. The figure graphs the unconditional average log distance between a firm and its lender against the year the relationship began. Data from the 1993 National Survey of Small Business Finance is graphed as triangles, while data from the 1988 survey is graphed as diamonds.

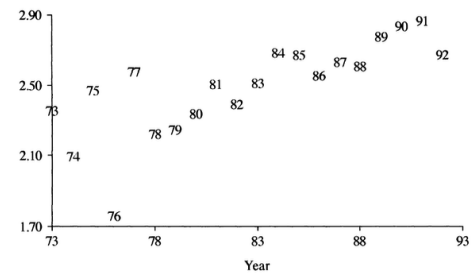


Figure 2. Changes in the estimated distance to lender over time. Estimated log distance between a firm and its lender across time is based on the estimates from Table IV (column IV). All variables except the year in which the relationship began are set equal to the sample mean. In the regression, the year variable is replaced by a set of year dummy variables. This allows the functional form of the relationship between distance and year to have any shape.

5.5 Table 5 and Figure 3: communication becomes impersonal

Table V
Determinants of the Method Used to Communicate with Lender: In Person, by Phone, or by Mail

The table contains estimates from an ordered logit model. The dependent variable denotes the predominant method for communicating with the lender. The sample includes only relationships beginning since 1973. Each observation represents a firm-lender pair. Firms that borrow from multiple lenders or that borrow from a given lender but through multiple loan types (e.g., a line of credit and a mortgage) will generate multiple observations. The data are taken from the 1993 National Survey of Small Business Finance. Models are described as follows: I. The dependent variable is coded one if in person, two if by phone, and three if by mail. Estimation is multinomial ordered logit. II. The log of one plus distance (the dependent variable in Table II) is added as an explanatory variable to the specification in column I. Estimation is multinomial ordered logit. III. The dependent variable is coded zero if in person and one otherwise. Estimation is logit.

Independent Variables	Models		
	I	II	III
Firm's age	0.005 ⁵ (0.002)	0.005 ⁵ (0.002)	0.006 ⁵ (0.003)
Year relation started	0.046 ¹ (0.007)	0.031 ¹ (0.007)	0.043 ¹ (0.007)
Lender is bank	-1.226 ¹ (0.079)	-0.875 ¹ (0.083)	-1.583 ¹ (0.099)
Lender is a nonfinancial firm	-0.432 ¹ (0.122)	-0.303 ⁵ (0.126)	-0.624 ¹ (0.161)
Lender provides a checking account	-1.637 ¹ (0.077)	-1.243 ¹ (0.081)	-1.403 ¹ (0.082)
Lender is firm's first lender	-0.215 ¹ (0.064)	-0.189 ¹ (0.066)	-0.290 ¹ (0.075)
Loan is collateralized	0.306 ¹ (0.087)	0.319 ¹ (0.091)	0.279 ¹ (0.094)
Firm had records for filling out survey	0.088 (0.057)	0.051 (0.059)	0.116 ¹⁰ (0.066)
Firm has credit card	0.048 (0.057)	0.032 (0.059)	0.128 ¹⁰ (0.067)
Ownership share of largest owner (%)	-0.214 ⁵ (0.101)	-0.107 (0.105)	-0.507 ¹ (0.119)
Corporation (1 = yes)	0.093 (0.072)	0.078 (0.075)	0.172 ⁵ (0.082)
Franchise (1 = yes)	-0.028 (0.120)	-0.187 (0.127)	0.076 (0.142)
Owner managed	-0.017 (0.067)	0.006 (0.069)	-0.134 ¹⁰ (0.079)
Owner's age when relation began	0.004 (0.004)	0.005 (0.004)	0.006 (0.004)
Owner's experience when relation began	-0.003 (0.004)	-0.003 (0.004)	-0.005 (0.004)
Sales area regional (1 = yes)	-0.024 (0.064)	-0.062 (0.066)	0.089 (0.073)
Sales area national (1 = yes)	0.382 ¹ (0.079)	0.202 ⁵ (0.082)	0.740 ¹ (0.094)
Firm in MSA (1 = yes)	0.302 ¹ (0.076)	0.441 ¹ (0.080)	0.317 ¹ (0.086)
Herfindahl index > 1,800	-0.076 (0.060)	-0.097 (0.062)	-0.162 ⁵ (0.070)
Log(1 + distance from lender)		0.436 ¹ (0.019)	
Pseudo R ²	0.216	0.265	0.284
Number of observations	5,945	5,945	5,945

¹⁰, ⁵, ¹ Significance at the 10%, 5%, and 1% levels, respectively.

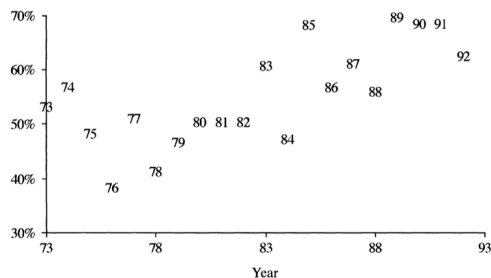


Figure 3. Changes in the estimated method of communicating with lender over time. The estimated probability of using nonpersonal communication (phone or mail) as the most frequent method of communicating with a lender as a function of the year the lending relationship started is graphed. The alternative method of communication is in person. The probability is calculated for a new firm that was both borrowing and had a checking account from a bank. The probabilities are based on the multinomial logit model (see Table V, column III), except the year the relationship started variable was replaced with a series of year dummy variables.

- **Read:** Table 5 shows that newer lending relationships are less likely to be conducted in person, and Figure 3 visualizes the fitted trend.
- **Key numbers.**
 - Probability of in-person communication falls from roughly **59%** for 1973-start relationships to **36%** for 1993-start relationships after controls.
 - After controlling for actual distance, it still falls from roughly **52%** to **37%**.
- **Punchline.** The information revolution changed *how* firms and lenders interact, not just how far apart they are.

5.6 Table 6, Table 7, Figure 4, and Figure 5: why consolidation is not the answer, and productivity is

- **Read:** These exhibits test candidate mechanisms.
- **Figure 4.** The number of banks falls mainly from the mid-1980s onward, but branch counts rise or remain much more stable.

Table VI
Determinants of Distance to the Firm's Lenders: Includes Changes in the Lending Environment
The table contains regressions where the dependent variable is the log of one plus the distance to the firm's lender. The regression contains all the variables from Table IV, although only a subset is reported. In most cases, the variables in the table vary across both time and census region. The sample includes only relationships beginning since 1973. Each observation represents a firm-lender pair. Firms that borrow from multiple lenders or that borrow from a given lender but through multiple loan types (e.g., a line of credit and a mortgage) will generate multiple observations. The data are taken from the 1993 National Survey of Small Business Finance. Column V is based only on observations where the lender is a bank.

Independent Variables	Models						
	I	II	III	IV	V	VI	VII
Log(banks/1,000 square miles)	-0.084 ¹ (0.023)						
Log(branches/1,000 square miles)		-0.027 ¹⁰ (0.020)		-0.008 (0.027)	-0.004 (0.031)	-0.018 (0.031)	-0.007 (0.027)
Loan losses/total loans (%)			-0.004 (0.044)				
Log(bank employees/real loans)				-0.185 (0.125)	-0.155 (0.133)		-0.189 (0.131)
Log(bank employees/real GDP)				-0.474 ¹ (0.181)	-0.483 ¹ (0.201)		-0.504 ¹ (0.185)
Log(bank employees/real loans) (based on national numbers)							0.215 (0.472)
Log(bank employees/real GDP) (based on national numbers)							0.655 (0.885)
Log(bank employees)							-0.531 ¹ (0.278)
Log(real GDP)							0.457 ¹ (0.183)
Log(real loans)							0.116 (0.144)
Year relation started	0.038 ¹ (0.005)	0.038 ¹ (0.005)	0.037 ¹ (0.006)	0.014 (0.011)	0.003 (0.012)	0.016 (0.011)	0.043 (0.037)
R ²	0.320	0.318	0.318	0.314	0.166	0.314	0.314
Number of observations	5,974	5,974	5,974	5,662	3,542	5,974	5,662

¹⁰, ¹, ⁵, ¹ Significance at the 10%, 5%, and 1% levels, respectively.

Table VII
Determinants of the Method Used to Communicate with Lender: Includes Changes in the Lending Environment

Estimates are from a logit model. The dependent variable denotes the predominant method for communicating with the lender (zero if in person and one if by phone or by mail). The models contain the variables from Table V, although only a subset is reported. Each observation represents a firm-lender pair. Firms that borrow from multiple lenders or that borrow from a given lender but through multiple loan types (e.g., line of credit and mortgage) will generate multiple observations. The data are taken from the 1993 National Survey of Small Business Finance. Column V is based only on observations where the lender is a bank.

Independent Variables	Models						
	I	II	III	IV	V	VI	VII
Log(banks/1,000 square miles)	-0.100 ¹ (0.036)						
Log(branches/1,000 square miles)		0.007 (0.031)		0.037 (0.042)	0.052 (0.049)	-0.001 (0.048)	0.035 (0.043)
Loan losses/total loans			-0.038 (0.059)				
Log(bank employees/real loans)				-0.641 ¹ (0.196)	-0.536 ¹ (0.225)		-0.681 ¹ (0.303)
Log(bank employees/real GDP)				-0.913 ¹ (0.284)	-0.951 ¹ (0.322)		-0.928 ¹ (0.285)
Log(bank employees/real loans) (based on national numbers)							0.456 (0.865)
Log(bank employees/real GDP) (based on national numbers)							0.394 (2.187)
Log(bank employees)							-1.130 ¹ (0.422)
Log(real GDP)							0.443 ¹ (0.245)
Log(real loans)							0.459 ¹ (0.225)
Year relation started	0.042 ¹ (0.007)	0.043 ¹ (0.007)	0.049 ¹ (0.008)	-0.002 (0.017)	-0.002 (0.018)		0.008 (0.140)
R ²	0.285	0.284	0.216	0.289	0.143	0.289	0.289
Number of observations	5,945	5,945	5,945	5,633	3,517	5,633	5,633

¹ and ² Significance at the 5% and 1% levels, respectively.

- **Distance and density.** Bank density has some effect on distance, but it explains only a small fraction of the secular rise.
- **Loan losses.** Loan losses do not explain the trends.
- **Figure 5.** Bank employment relative to loans or regional GDP declines over time, consistent with higher lender productivity.
- **Key quantitative result.** A 10% decrease in bank employees relative to real regional GDP raises lender distance by about **4.7%**, and the year trend largely disappears.
- **Punchline.** Information technology and labor-saving change in lending production fit the data much better than merger or lax-standards stories.

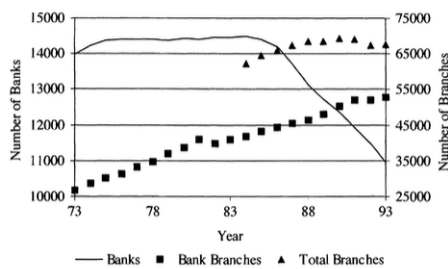


Figure 4. Number of banks and bank branches. The number of banks and bank branches is taken from the Federal Deposit Insurance Corporation (FDIC) web site. The number of banks is graphed along the left axis as a line and is stable until 1984 when it begins to decline. The number of branches is graphed along the right axis. The graph contains both the number of bank branches and the number of bank plus S&L (total) branches. The data on S&L branches is available only since 1984.

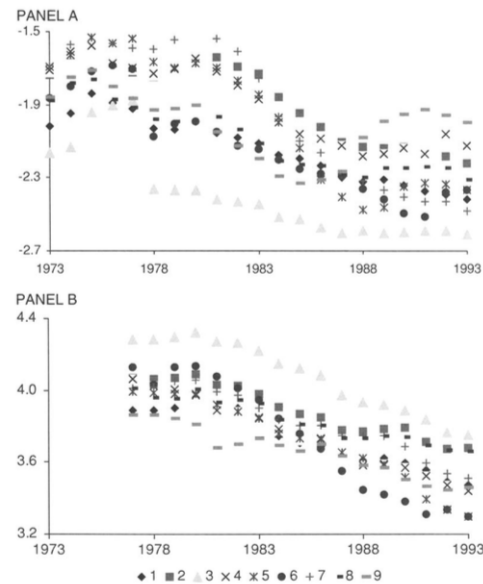


Figure 5. Total bank employment relative to total real bank lending and total bank employment relative to real gross region product. Panel A is the graph by census region (1-9) and year of the log of total bank employment divided by total real bank lending (in \$1,000s) against time. Both numbers are from the Federal Deposit Insurance Corporation (FDIC). Regional numbers are the sum of the numbers for each state in the region. Panel B is the graph by census region and year of the log of total bank employment divided by real gross region product. Gross state product data is from the Bureau of Economic Analysis (BEA), Commerce Department. (The FDIC numbers are from <http://www2.fdic.gov/hsob/> and the Commerce department numbers are from <http://www.bea.doc.gov/bea/regional/gsp/gsp1ist.htm>.) Gross state product is only available since 1977 and is reported in millions of dollars. Regional numbers are the sum of the numbers for each state in the region.

5.7 Table 8 and Table 9: credit-quality consequences

- **Read:** These tables ask whether informational transparency, measured by predicted distance or predicted communication, predicts better credit outcomes and whether that predictive power changes over time.
- **Rates.** More transparent firms pay lower rates, but the interaction with time is not significant.
- **Approval.** More transparent firms are much more likely to get approved, but this advantage weakens over time.
- **Punchline.** Distant borrowing used to signal very high borrower quality; over time, that signal weakens because technology lets more ordinary firms access distant credit.

Table VIII

Determinants of Interest Rate on Firm's Most Recent Loan

The dependent variable is the rate on the most recent loan. The estimates in columns III and VI are identical to those in columns II and V, except the standard errors have been estimated by bootstrapping (1,000 repetitions). The models also include a dummy variable for whether the firm is a corporation, whether the loan is collateralized, and whether the firm or the firm's owner has had a delinquency in the last three years. The data are taken from the 1993 National Survey of Small Business Finance.

Independent Variables	Models					
	I	II	III	IV	V	VI
Maturity matched treasury rate	0.367 ¹ (0.029)	0.367 ¹ (0.029)	0.367 ¹ (0.029)	0.369 ¹ (0.029)	0.370 ¹ (0.029)	0.370 ¹ (0.029)
Log(firm's assets)	-0.243 ¹ (0.021)	-0.243 ¹ (0.021)	-0.243 ¹ (0.021)	-0.242 ¹ (0.021)	-0.242 ¹ (0.021)	-0.242 ¹ (0.022)
Outside debt/assets	-0.107 (0.067)	-0.107 (0.067)	-0.107 (0.080)	-0.104 (0.067)	-0.103 (0.067)	-0.104 (0.079)
Log(1 + firm age)	-0.024 (0.044)	-0.032 (0.048)	-0.032 (0.049)	-0.015 (0.044)	-0.012 (0.047)	-0.012 (0.050)
Loan has floating rate (1 = yes)	0.030 (0.074)	0.030 (0.075)	0.030 (0.071)	0.032 (0.075)	0.032 (0.075)	0.032 (0.072)
Lender is a bank (1 = yes)	-0.565 ¹ (0.099)	-0.567 ¹ (0.099)	-0.567 ¹ (0.148)	-0.557 ¹ (0.099)	-0.556 ¹ (0.099)	-0.556 ¹ (0.150)
Lender is nonfinancial firm	-0.460 ⁵ (0.230)	-0.459 ⁵ (0.230)	-0.459 (0.485)	-0.443 ⁵ (0.230)	-0.444 ⁵ (0.230)	-0.444 (0.484)
Number of financial lenders	0.075 ¹ (0.021)	0.076 ¹ (0.021)	0.076 ¹ (0.018)	0.073 ¹ (0.021)	0.073 ¹ (0.021)	0.073 ¹ (0.018)
Predicted distance from lender	-0.546 ¹ (0.179)	-0.528 ¹ (0.184)	-0.528 ⁵ (0.001)			
Predicted distance * year relation started		-0.001 (0.004)	-0.001 (0.003)			
Predicted method used to communicate with lender				-1.328 ¹ (0.418)	-1.398 ¹ (0.542)	-1.398 ¹ (0.546)
Predicted method * year relation started					0.005 (0.024)	0.005 (0.021)
R ²	0.174	0.175	0.175	0.175	0.175	0.175
Number of observations	3,523	3,523	3,523	3,523	3,523	3,523

⁵ and ¹ Significance at the 5% and 1% levels, respectively.

Table IX

Probability of Loan Approval on Most Recent Loan

The table contains estimates from a logit model. The independent variable is one if the firm was approved for a loan and zero otherwise. The sample includes both those firms that applied for a loan as well as those firms that did not apply for a loan because they expected to be turned down. The last group is coded as being turned down for a loan. Only loans that were applied for (or considered) in the last three years are included. The estimates in columns III and VI are identical to those in columns II and V, but the standard errors have been estimated by bootstrapping (1,000 repetitions). The models also include a dummy variable for whether the firm is a corporation, whether the loan is collateralized, and whether the firm or the firm's owner has had a delinquency in the last three years. The data are taken from the 1993 National Survey of Small Business Finance.

Independent Variables	Models					
	I	II	III	IV	V	VI
Log(firm's assets)	0.350 ¹ (0.026)	0.349 ¹ (0.026)	0.349 ¹ (0.026)	0.352 ¹ (0.026)	0.349 ¹ (0.026)	0.349 ¹ (0.025)
Return on assets (profits/assets)	0.375 ¹ (0.080)	0.386 ¹ (0.080)	0.386 ¹ (0.080)	0.382 ¹ (0.079)	0.381 ¹ (0.080)	0.381 ¹ (0.080)
Outside debt/assets	0.005 (0.015)	0.005 (0.015)	0.005 (0.020)	0.004 (0.015)	0.004 (0.015)	0.004 (0.021)
Corporation (1 = yes)	-0.034 (0.104)	-0.028 (0.104)	-0.028 (0.115)	-0.037 (0.105)	-0.036 (0.105)	-0.036 (0.111)
Firm age	0.008 (0.006)	0.008 (0.006)	0.008 (0.005)	0.007 (0.005)	0.007 (0.005)	0.007 (0.005)
Length of longest relationship	0.012 (0.009)	0.004 (0.010)	0.004 (0.010)	0.012 (0.009)	0.004 (0.010)	0.004 (0.010)
Debt from financial service provider (%)	1.590 ¹ (0.107)	1.563 ¹ (0.108)	1.563 ¹ (0.115)	1.584 ¹ (0.107)	1.554 ¹ (0.108)	1.554 ¹ (0.113)
Number of financial lenders	0.021 (0.029)	0.028 (0.029)	0.028 (0.029)	0.020 (0.029)	0.029 (0.029)	0.029 (0.030)
Herfindahl index > 1,800	0.187 ⁵ (0.080)	0.184 ⁵ (0.080)	0.184 ⁵ (0.078)	0.191 ⁵ (0.080)	0.187 ⁵ (0.080)	0.187 ⁵ (0.079)
Predicted distance from lender	0.680 ¹ (0.257)	0.824 ¹ (0.269)	0.824 ¹ (0.297)			
Predicted distance * year relation started		-0.010 ¹⁰ (0.006)	-0.010 ¹⁰ (0.006)			
Predicted method used to communicate with lender				1.614 ¹ (0.638)	2.927 ¹ (0.928)	2.927 ¹ (0.967)
Predicted method * year relation started					-0.084 ⁵ (0.024)	-0.084 ⁵ (0.043)
Pseudo R ²	0.172	0.173	0.173	0.172	0.173	0.173
Number of observations	4,548	4,548	4,548	4,548	4,548	4,548

¹⁰, ⁵, ¹ Significance at the 10%, 5%, and 1% levels, respectively.

6 Short summary

- **Method.** Reduced-form cohort regressions on firm–lender relationships, using distance and communication measures, plus extensive robustness checks and later credit-quality regressions based on predicted transparency.
- **Identification.** The paper does not rely on an external instrument. Its credibility comes from rich controls, 1988/1993 survey comparison, within-firm evidence, rejection of alternative explanations, and region-specific lender-productivity measures.
- **Application.** U.S. small business lending relationships formed from 1973 to 1993.

- **Main result.** Small business lending becomes much less local and much less personal over time. The most convincing explanation is the information revolution: more abundant hard information and more productive lender screening and monitoring technologies.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that small firms in the United States increasingly borrowed from **more distant lenders** between the 1970s and early 1990s.
- It also establishes that firms and lenders increasingly interacted in **impersonal ways**, using phone and mail rather than face-to-face contact.
- These changes are not well explained by bank consolidation, changing borrower location, deposit-market deregulation, or declining loan quality.
- They are most consistent with a lending technology that relies less on local soft information and more on hard information, infomediaries, and automated evaluation and monitoring.

7.2 Economic intuition

- Historically, small business lending was local because information was soft, tacit, and expensive to transmit.
- As information technology improved, lenders could obtain timely hard information even at a distance. That changed the trade-off between lending locally and lending remotely.
- Once lenders can monitor from afar and intervene quickly when conditions deteriorate, they no longer need the same degree of geographic proximity.
- The result is a bigger effective credit market for small firms: more potential lenders, more competition, and broader access to finance.

7.3 Takeaway

This paper's main lesson is that distance stopped being as central to small business lending as classic relationship-lending models would suggest. The information revolution did not eliminate the value of relationships, but it made them less dependent on physical proximity. That insight matters for how we think about bank competition, financial geography, local shocks, and the real effects of improvements in information technology.